Telco Customer Churn Dataset

Dataset Overview:

This dataset contains customer-level information from a telecommunications company.The goal of this task is to analyze customer behavior and apply deep learning techniques to predict customer churn (whether a customer is likely to leave the service. perform data analysis and preprocessing, feature encoding, model building, and evaluation using deep learning models.

1. Customer Identification

- customerID: Unique identifier assigned to each customer.

2. Demographic Information

- gender: Gender of the customer.
- SeniorCitizen: Indicates whether the customer is a senior citizen (1 = Yes, 0 = No).
- Partner: Indicates whether the customer has a partner.
- Dependents: Indicates whether the customer has dependents.

3. Account Information

- tenure: Number of months the customer has been with the company.
- Contract: Type of contract the customer has.
- PaperlessBilling: Indicates whether the customer uses paperless billing.
- PaymentMethod: Payment method used by the customer to pay bills.

4. Phone Services

- PhoneService: Indicates whether the customer subscribes to phone service.
- MultipleLines: Indicates whether the customer has multiple phone lines.

5. Internet and Add-on Services

- InternetService: Type of internet service subscribed by the customer.
- OnlineSecurity: Indicates whether online security service is enabled.
- OnlineBackup: Indicates whether online backup service is enabled.
- DeviceProtection: Indicates whether device protection service is enabled.
- TechSupport: Indicates whether technical support service is enabled.
- StreamingTV: Indicates whether TV streaming service is enabled.
- StreamingMovies: Indicates whether movie streaming service is enabled.

6. Billing Information

- MonthlyCharges: Monthly amount charged to the customer.
- TotalCharges: Total amount charged to the customer over the entire tenure.

7. Target Variable

- Churn: Target variable indicating whether the customer left the company.